Semantic-Guided Imitation Learning with History-Conditioned Feedback

***Abstract*—**

***Index Terms*—Imitation learning, semantic representations, visuomotor control, residual feedback.**

I. Introduction

Reliable execution of sequential manipulation tasks is essential for practical robotic applications but remains challenging [1]. Sequential manipulation tasks require a robot to achieve an overall goal through a sequence of interrelated operations [2]. Such tasks arise in practical applications ranging from household cleaning, where a tablecloth is lifted to clear debris beneath it and then restored, to surgery, where tissue is retracted to expose the target region before resection [3]. A key challenge of these tasks is that preceding interactions change the conditions under which subsequent operations can be performed. For example, compared with simple pick-and-place tasks involving visible and accessible targets, manipulation under occlusion requires the robot to assess whether prior actions have made the target visible and accessible or further manipulation is needed [4], [5]. Consequently, reliable execution requires a policy that adapts subsequent actions to interaction outcomes observed through visual feedback.

Visuomotor imitation learning offers a practical approach to sequential manipulation by reducing the need for explicit interaction modeling, but reliable execution remains difficult with limited demonstrations. It learns a policy that maps visual and proprioceptive observations to actions using expert demonstrations. For example, Action Chunking with Transformers (ACT) [6] has been used in Mobile ALOHA to learn coordinated mobile manipulation, including opening a cabinet and storing a cooking pot [7]. Similarly, Diffusion Policy [8] has been adopted in the Universal Manipulation Interface (UMI) framework to learn bimanual cloth folding and multi-step dish washing [9]. However, execution errors can lead a policy into states poorly covered by limited demonstrations, where its predicted actions may no longer suit the actual task conditions, compromising subsequent operations and task completion [10].

Prior work has focused on improving the robustness of visuomotor policies to execution deviations by expanding data coverage, improving visual representations, and adapting actions during execution. Ross et al. [10] expanded the training dataset with expert action labels for states visited by the learned policy, while Laskey et al. [11] injected noise into expert actions during demonstration collection to obtain examples of recovery behavior. Although data coverage expansion improves coverage of states arising from execution deviations, it requires additional expert supervision or modifications to demonstration collection, limiting its applicability when only a fixed demonstration set is available. Zhu et al. introduced object-centric visual representations for robust visuomotor imitation [12] and incorporated 3D object representations to improve generalization across backgrounds, viewpoints, and object instances [13]. Although these methods improve generalization to visual variations, robust object representations alone do not ensure that subsequent actions remain appropriate when execution deviates from demonstrations. Ankile et al. [14] trained a residual policy through reinforcement learning to correct the actions of a frozen imitation policy, while Sendai et al. [15] learned corrections from demonstrations using updated observations and base-policy context to refine actions during chunk execution. However, reinforcement-based correction requires additional environment interaction, whereas corrections based on updated observations and base-policy context may be insufficient when appropriate adjustments depend on preceding actions and their observed effects.

In this work, we develop a theory-guided visuomotor imitation learning framework that combines task-relevant representation learning with history-conditioned feedback to support sequential manipulation from limited demonstrations. Our analysis derives conditions linking imitation error and representation quality to task-failure bounds. It further identifies when updated observations and execution history provide additional information for predicting residual action errors, and when learned corrections can realize this benefit. Guided by this analysis, we supervise policy query tokens with task descriptors derived from semantic masks, encouraging the policy to retain information relevant to changing manipulation conditions. We then train a residual model that combines updated observations, the supervised query features, and action–response histories to adjust pending actions in the current plan according to observed interaction outcomes. Both the base policy and the residual model are trained from the same fixed demonstration set, without additional robot interaction for policy training. Simulation and real-world experiments demonstrate the effectiveness of the proposed framework in sequential manipulation tasks.

II. Related Work

III. Problem Formulation and Theoretical Analysis

This section studies conditions for improving task performance from limited demonstrations. Starting from the relationship between imitation error and task success, we examine two paths for improvement: preserving information needed to distinguish the task consequences of different actions, and using new observations and execution history to produce effective corrections.

We first establish conditions under which action error bounds task-failure probability and explain why improved coverage in representation space alone does not guarantee better task performance. We then examine how semantic guidance can preserve task-relevant information and, under consistent expert selection, control imitation-error transfer through expert-action reconstruction. Finally, we characterize the predictive value of updated observations and history, and identify when learned residual corrections reduce imitation error and tighten the task-performance bound.

A. Visuomotor Imitation Learning

We study offline visuomotor imitation learning from a finite, fixed set of robot demonstrations, for manipulation tasks whose completion can be assessed from visual history. The policy generates control commands from visual and proprioceptive observations and available execution history. Training is based on this fixed demonstration dataset, with demonstrated actions providing the primary imitation target, whereas deployment performance is measured by the probability of task completion over a finite horizon.

1) System and Observations: Consider a robotic system governed by a single decision policy that performs a given manipulation task through the joint actions of its actuators. The joint robot-environment state $s_t \in \mathcal{S}$ evolves under the command $u_t \in \mathcal{A}$ and gives rise to observation o_t. The observation $o_t = (I_t^{1:V}, q_t)$ comprises multi-view images $I_t^{1:V} = (I_t^1, \dots, I_t^V)$ and proprioceptive information q_t, where $V \geq 1$ is the number of cameras and I_t^v is the image from the vth view. Over a finite horizon of T control steps, this interaction is represented by the following partially observed controlled process:

$$\begin{aligned} s_0 &\sim p_0, \\ s_{t+1} &\sim P(\cdot \mid s_t, u_t), \quad 0 \leq t < T, \\ o_t &\sim O(\cdot \mid s_t), \quad 0 \leq t \leq T. \end{aligned} \tag{1}$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A} \subseteq \mathbb{R}^{d_u}$ is the admissible command set specified by the chosen robot control interface, d_u is the command dimension, p_0 is the initial-state distribution, and P and O are the transition and observation kernels, respectively.

2) Offline Demonstrations and Policy: The policy learns from a fixed demonstration set $\mathcal{D} = \{\zeta_i\}_{i=1}^N$ for the same manipulation task. The ith demonstration trajectory ζ_i records observations and demonstrated control commands:

$$\zeta_i = (o_0^i, u_0^{\mathrm{E},i}, o_1^i, \dots, u_{T_i-1}^{\mathrm{E},i}, o_{T_i}^i). \tag{2}$$

Here, N is the finite number of demonstrations, $T_i \geq 1$ is the number of control steps in the ith trajectory, and the superscript E denotes demonstrated commands.

The information available to the deployed policy π is represented by the history $\mathcal{H}_t$. This history includes observations $o_{0:t}$ obtained up to time t, previously executed commands $u_{0:t-1}$, and other runtime records before the current decision. The policy defines a distribution over admissible control commands, with $u_t \sim \pi(\cdot \mid \mathcal{H}_t)$. The command history is empty at $t = 0$.

3) Learning and Task Objectives: For the following analysis, we consider deterministic policies acting on the available history, with π_{E} denoting the expert policy. Let ν denote the population distribution underlying the demonstrations, where $(t, h) \sim \nu$ specifies a control step and its available history. We define the population imitation risk as the expected squared Euclidean deviation from the expert's action:

$$R_\nu(\pi) := \mathbb{E}_{(t,h)\sim\nu} \|\pi(h) - \pi_{\mathrm{E}}(h)\|^2. \tag{3}$$

Training seeks to reduce this risk through an empirical imitation loss on the finite demonstrations; the specific training losses are defined in Section IV.

The deployment objective is to complete the task within the finite horizon. The measurable task-completion function $\Phi : \mathcal{S}^{T+1} \to \{0, 1\}$ equals 1 for state trajectories satisfying the task's terminal or temporal requirements and 0 otherwise. This defines the completion indicator Ψ and the task-completion probability $J(\pi)$ of the deployed policy π:

$$\Psi = \Phi(s_{0:T}), \qquad J(\pi) = \mathbb{P}_\pi(\Psi = 1). \tag{4}$$

Here, $\mathbb{P}_\pi$ is the trajectory probability law induced jointly by the system model and the complete deployed policy π. Learning aims to reduce $R_\nu(\pi)$, whereas deployment aims to increase $J(\pi)$. A lower imitation risk does not by itself guarantee a higher task-completion probability; the following analysis examines conditions relating these objectives.

4) Visual Assessability: Under a fixed camera configuration and evaluation horizon T, the evaluation domain $\Omega_{\text{eval}} \subseteq \mathcal{S}^{T+1} \times \mathcal{I}^{T+1}$ consists of prespecified pairs of physical trajectories and visual histories consistent with the system model and visibility conditions, where $\mathcal{I}$ is the space of multi-view image tuples $I_t^{1:V}$.

Assumption 1 (Visual assessability): There exists a measurable visual decision function $\mathcal{V} : \mathcal{I}^{T+1} \to \{0, 1\}$ such that, for every $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$,

$$\Phi(s_{0:T}) = \mathcal{V}(I_{0:T}^{1:V}). \tag{5}$$

Assumption 1 specifies visual identifiability of completion within the evaluation domain, without requiring full-state observability. The subsequent analysis considers policy executions and counterfactual action comparisons for which $(s_{0:T}, I_{0:T}^{1:V}) \in \Omega_{\text{eval}}$ almost surely.

B. From Imitation Error to Task Performance

Whether action error bounds task failure depends on where deviations occur and their subsequent task consequences. We establish this connection and show that transferring imitation risk from demonstrations to deployment depends jointly on representation-level coverage and conditional action-error discrepancy.

1) Task-Performance Difference Between Policies: We compare a policy π with a reference policy π_0 through their task-failure probabilities and express the difference as a sum of expected single-step action consequences. Each consequence is defined by comparing alternative actions at the same history while following π_0 afterward [16], [17]. This decomposition

provides the basis for relating imitation error to task performance.

We consider deterministic policies acting on the available history. Given a history $\mathcal{H}_t = h$ and a candidate action $a \in \mathcal{A}$, define the action-value function as the probability of task failure ($\Psi = 0$):

$$Q_t^{\pi_0}(h, a) := \mathbb{P}^{a,\pi_0}(\Psi = 0 \mid \mathcal{H}_t = h). \tag{6}$$

Here, $\mathbb{P}^{a,\pi_0}$ denotes the probability law obtained by taking action a at time t and following π_0 afterward.

When the current action is also selected by π_0, this reduces to the value function of π_0, $V_t^{\pi_0}(h) := Q_t^{\pi_0}\big(h, \pi_0(h)\big)$. The effect of choosing a instead of $\pi_0(h)$ is therefore measured by

$$A_t^{\pi_0}(h, a) := Q_t^{\pi_0}(h, a) - V_t^{\pi_0}(h). \tag{7}$$

Thus, $A_t^{\pi_0}(h, a) > 0$ indicates an increase in failure probability, whereas $A_t^{\pi_0}(h, a) < 0$ indicates a decrease.

To relate these action-level consequences to differences in task performance between policies, we require the following common comparison conditions.

Assumption 2 (Common task and environment): The compared policies share the same initial distribution and task-completion criterion. Given a history h and an action a, the next available history follows a conditional distribution that does not otherwise depend on the policy. The reference policy π_0 is defined on all histories involved in the comparison.

Let $F(\pi) = 1 - J(\pi)$ denote the task-failure probability of policy π. To evaluate π relative to π_0, we set $a = \pi(h)$ and aggregate the resulting action consequences along trajectories generated by π.

Applying the finite-horizon performance-difference identity [16]–[18] to these one-step action effects gives

$$F(\pi) - F(\pi_0) = \mathbb{E}_\pi\left[\sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))\right]. \tag{8}$$

Here, $\mathbb{E}_\pi$ denotes expectation over trajectories generated by π. Equation (8) evaluates action effects at the situations actually visited by the evaluated policy π. Its proof is given in Appendix A-A. This relation allows task-performance differences to be bounded through the consequences of action deviations.

2) Task-Failure Bounds from Imitation Risk: Taking the expert policy as the reference, we show that the imitation risk $R_\nu(\pi)$ bounds task-failure probability $F(\pi)$ when demonstrations sufficiently cover deployment histories and action deviations have bounded task consequences.

Imitation risk is evaluated under the demonstration distribution ν, but deployment histories may follow a different distribution because actions depend on π. Let μ_π^t denote the distribution of $(t, \mathcal{H}_t)$ at control step t under π. The distribution $\bar{\mu}_\pi = T^{-1}\sum_{t=0}^{T-1}\mu_\pi^t$ corresponds to choosing a control step uniformly and observing its history.

Assumption 3 (Demonstration coverage):

For the evaluated policy π, there exists a uniform constant $1 \le C < \infty$ such that

$$\bar{\mu}_\pi(\mathcal{B}) \le C\,\nu(\mathcal{B})$$

for every measurable set $\mathcal{B}$ of pairs (t, h).

This condition excludes deployment situations absent from the reference distribution and limits how much their probabilities can be underestimated.

Assumption 4 (Action sensitivity): Let A_t^{E} denote the action effect with the expert policy as the reference. For a fixed action scale, there exists a uniform constant $0 \le L < \infty$ such that

$$|A_t^{\mathrm{E}}(h, a)| \le L\|a - \pi_{\mathrm{E}}(h)\|$$

for all relevant (t, h, a).
This assumption allows action deviations to be translated into bounds on changes in task-failure probability.

Following the connection between deployment imitation error and task cost [10], Assumptions 3 and 4, together with the performance-difference identity, yield the following bound (Appendix A-B):

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{CR_\nu(\pi)}. \tag{9}$$

Here, C accounts for the shift from demonstration to deployment distributions, while L measures the sensitivity of task outcomes to action deviations. With other quantities fixed, a smaller $R_\nu(\pi)$ yields a tighter upper bound on task-failure probability.

This bound identifies coverage and action sensitivity as key conditions for translating imitation accuracy into task-performance guarantees. We next examine the limitations of complete-history coverage and the role of representations in risk transfer.

3) Imitation-Risk Transfer Through Representations: To address the limitations of complete-history coverage, we analyze imitation-risk transfer in representation space. We derive a bound showing that compression can relax coverage requirements, while differences in conditional imitation error must also be accounted for.

Assumption 3 can fail even after an arbitrarily small action deviation. For $T \ge 2$, suppose the expert always executes $u_0 = 0$, whereas the learned policy executes an admissible $u_0 = \varepsilon \ne 0$. Histories recording $u_0 = \varepsilon$ then occur under the learned policy but have zero probability under the expert demonstration distribution. No finite coverage constant C can satisfy the assumption.

A representation can merge such distinctions, potentially relaxing coverage requirements, but may also discard information needed to distinguish action consequences. This tension between distributional alignment and information loss under a noninvertible representation also appears in domain-adaptation analyses [19], [20].

For a fixed policy, expert, and common measurable causal encoder $Z = \phi(t, \mathcal{H}_t)$ retaining the time index, let μ_Z and ν_Z be the representation distributions induced by $\bar{\mu}_\pi$ and ν. In place of Assumption 3, suppose $\mu_Z(\mathcal{B}) \le C_\phi \nu_Z(\mathcal{B})$ for every measurable representation event $\mathcal{B}$, with $1 \le C_\phi < \infty$. Even at the same representation, the two distributions may contain different histories. Let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$ and define their conditional-error discrepancy as $D_\phi = \mathbb{E}_{z\sim\mu_Z}|m_\mu(z) - m_\nu(z)|$, where $m_\mu(z) = \mathbb{E}_{\bar{\mu}_\pi}[e(h)^2 \mid Z = z]$ and $m_\nu(z) = \mathbb{E}_\nu[e(h)^2 \mid Z = z]$. With square-integrable action error under

both distributions and measurable versions of these conditional means,

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C_\phi R_\nu(\pi) + D_\phi. \tag{10}$$

If Assumption 4 also holds, the performance-difference identity gives

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + D_\phi}. \tag{11}$$

Appendix A-C provides the proof. With imitation risk and action sensitivity fixed, (11) shows that the effect of a representation on the bound depends jointly on the coverage factor C_ϕ and the conditional error discrepancy D_ϕ. This motivates seeking representations that reduce C_ϕ while controlling D_ϕ, so that their combined contribution $C_\phi R_\nu(\pi) + D_\phi$ decreases.

C. Task-Relevant Information Preservation

We seek conditions under which a representation can improve coverage while controlling the conditional imitation-error discrepancy. We examine semantic guidance as a way to retain task-relevant information, then connect the accuracy of that information to expert-action reconstruction and a bound on D_ϕ.

1) Semantic-Guided Compression and Coverage: Semantic guidance can encourage an encoder to merge appearance differences while retaining object-level structure relevant to action selection. This motivates a representation design, not an automatic coverage guarantee. As in Section III-B3, compression can relax coverage requirements by making some history distinctions irrelevant to representation events. Appending a deterministic semantic map while retaining the original RGB input does not itself change the best coverage constant relative to RGB, since projection recovers that input. The relevant object is the learned decision representation $Z = \phi(t, \mathcal{H}_t)$, not the number of input modalities.

Improved coverage alone does not ensure a tighter bound; the error associated with merging histories must also be accounted for. A sufficient route is to retain information from which a common rule can reconstruct expert actions under both demonstration and deployment distributions. This emphasis on decision-relevant information is consistent with information-state and behaviorally grounded representation analyses [21], [22]. We next formulate this requirement using object attributes and spatial relationships.

2) Task Information and Expert-Action Reconstruction: We examine when retained task information is sufficient to approximate expert choices. The reconstruction error depends on both what the selected information fails to explain and how accurately it is estimated.

Fix the evaluated policy π and set $\mu = \bar{\mu}_\pi$, with ν the demonstration distribution. Write $\pi(h) = p(Z)$ for the same measurable policy readout under both distributions. We define a task descriptor vector $G \in \mathbb{R}^{d_G}$ whose components describe task-relevant object attributes and spatial relationships. Semantic masks provide a way to construct such quantities, including visibility and relative position. Let χ denote task-specified proprioceptive and execution context, selected in advance rather than identified with the unrestricted history. Suppose Z retains χ and supports an estimate $\widehat{G} \in \mathbb{R}^{d_G}$ of G. For $\rho \in \{\mu, \nu\}$, define $\delta_{G,\rho} = \mathbb{E}_\rho \|G - \widehat{G}\|$. These quantities must be defined under both distributions, with an explicit convention for unavailable components.

Assumption 5 (Descriptor-conditioned expert selection): There exists a common measurable rule g_{E} mapping descriptors and context to $\mathcal{A}$ such that, for $\rho \in \{\mu, \nu\}$,

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - g_{\mathrm{E}}(G, \chi)\| \le \epsilon_{\mathrm{sel},\rho}. \tag{12}$$

For a finite constant $K_{\mathrm{E}} \ge 0$,

$$\|g_{\mathrm{E}}(G, \chi) - g_{\mathrm{E}}(\widehat{G}, \chi)\| \le K_{\mathrm{E}} \|G - \widehat{G}\| \tag{13}$$

holds almost surely under both distributions, with finite $\epsilon_{\mathrm{sel},\rho}$ and $\delta_{G,\rho}$.

Here, $\epsilon_{\mathrm{sel},\rho}$ measures expert choices unexplained by (G, χ), while K_{E} controls sensitivity to estimation error. Task relevance alone does not imply this condition: equally successful actions may reflect different expert choices.

Define $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality yields

$$\mathbb{E}_\rho \|\pi_{\mathrm{E}}(h) - \widehat{a}_{\mathrm{E}}(Z)\| \le b_\rho := \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}} \delta_{G,\rho}. \tag{14}$$

Appendix B-A gives the derivation. Supervising descriptor readouts targets the estimation term, while the choice of descriptors and context determines how well expert actions can be explained.

3) Bounds on Conditional Imitation-Error Discrepancy: Expert-action reconstruction connects information preservation to the discrepancy in Section III-B3. Because both $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z)$ depend on the same representation, their squared difference is a common function under μ and ν. Expert reconstruction error bounds how far the actual conditional imitation errors depart from this function.

Assume a finite action-set diameter $B_{\mathcal{A}} = \sup_{a,a' \in \mathcal{A}} \|a - a'\|$ under a fixed action scale.

Proposition 1 (Conditional-error transfer through expert reconstruction): Under Assumption 5 and the representation-level coverage condition in Section III-B3, with b_ρ defined in (14),

$$D_\phi \le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu). \tag{15}$$

If Assumption 4 also holds,

$$F(\pi) \le F(\pi_{\mathrm{E}}) + TL\sqrt{C_\phi R_\nu(\pi) + 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu)}. \tag{16}$$

The proof is in Appendix B-B. The design objective is therefore to improve coverage while preserving information needed to reconstruct expert actions. For a fixed expert baseline, horizon, and action sensitivity, reducing the combined imitation and reconstruction terms tightens the task-failure bound.

D. Information-Conditioned Residual Corrections

Section III-C identifies conditions for preserving information needed to reconstruct expert actions. Such preservation does not eliminate the learned policy's imitation error, particularly when actions rely on outdated observations or limited execution history. We examine when additional observations and history make the remaining error more predictable, and when learned corrections realize this potential to reduce R_ν.

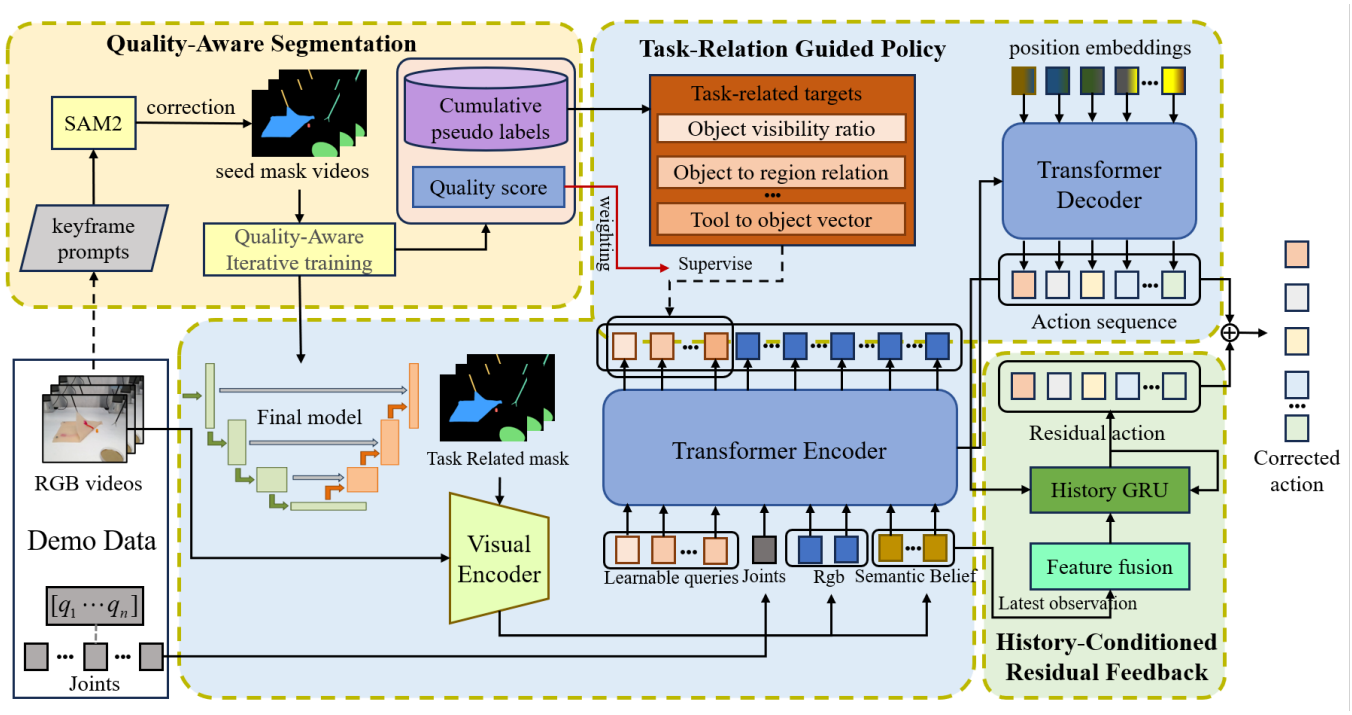


Fig. 1. Method overview. Quality-aware iterative annotation provides a frozen segmenter and quality-weighted descriptor labels. RGB, soft semantics, proprioception, and supervised queries jointly support action-chunk prediction. History-conditioned feedback uses updated observations and the current plan to correct unexecuted actions.

1) Residual Prediction of Action Error: We treat residual learning as prediction of the action deviation left by a fixed base policy. All comparisons use the same demonstration distribution ν, expert targets, target execution steps, and action scale. For base policy π_{b}, let $B = \pi_{\mathrm{b}}(\mathcal{H}_t)$ and define the residual target

$$E = \pi_{\mathrm{E}}(\mathcal{H}_t) - B. \tag{17}$$

Let $\mathcal{F}_{\mathrm{b}} \subseteq \mathcal{F}_{\mathrm{c}} \subseteq \mathcal{F}_{\mathrm{h}} \subseteq \sigma(t, \mathcal{H}_t)$ denote the information used to generate B, its augmentation with updated observations, and its further augmentation with execution history. The base action is $\mathcal{F}_{\mathrm{b}}$-measurable; all added information is available before the target action is executed. For any such information $\mathcal{F}$ and $\mathbb{E}_\nu\|E\|^2 < \infty$, the ideal residual predictor and its minimum squared error are

$$\begin{aligned} r^*_{\mathcal{F}} &= \mathbb{E}_\nu[E \mid \mathcal{F}], \\ R^*_\nu(\mathcal{F}) &= \mathbb{E}_\nu\|E - r^*_{\mathcal{F}}\|^2. \end{aligned} \tag{18}$$

This is an unrestricted square-integrable prediction benchmark, not an assumption that a network attains it or that $B + r^*_{\mathcal{F}}$ satisfies all action constraints. The derivation is in Appendix C-A.

2) Predictive Value of Additional Information: Additional information has strict predictive value when it changes the best estimate of the remaining action error. For $\mathcal{F}_0 \subseteq \mathcal{F}_1$,

$$R^*_\nu(\mathcal{F}_0) - R^*_\nu(\mathcal{F}_1) = \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2. \tag{19}$$

Applying this identity to $(\mathcal{F}_{\mathrm{b}}, \mathcal{F}_{\mathrm{c}})$ and $(\mathcal{F}_{\mathrm{c}}, \mathcal{F}_{\mathrm{h}})$ separates the potential contributions of updated observations and history. The gain is zero if the conditional mean is unchanged; more information alone does not imply a strict improvement. Appendix C-B gives the proof. These gains concern optimal prediction, not the performance of a particular history model.

3) Realized Corrections and Task-Performance Bounds: A learned correction improves imitation accuracy only when its prediction error is smaller than the error that the available information makes correctable. Let $r_{\mathcal{F}}$ be the square-integrable, $\mathcal{F}$-measurable residual actually applied, and let $\pi_{\mathcal{F}}(\mathcal{H}_t) = B + r_{\mathcal{F}}$. Define $\epsilon_{\mathrm{res}}(\mathcal{F}) = \mathbb{E}_\nu\|r_{\mathcal{F}} - r^*_{\mathcal{F}}\|^2$. Then

$$R_\nu(\pi_{\mathcal{F}}) = R_\nu(\pi_{\mathrm{b}}) - \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2 + \epsilon_{\mathrm{res}}(\mathcal{F}). \tag{20}$$

Thus, R_ν decreases if $\epsilon_{\mathrm{res}}(\mathcal{F}) < \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2$. The realization error includes model limitations, imperfect learning, and output constraints. Richer information improves the learned correction only if its predictive gain exceeds any increase in this error (Appendix C-C).

For admissible corrected actions, the task-performance implication follows from (11) under its coverage and action-sensitivity conditions. Corrections may change the deployment distribution, so the relevant quantity remains $C_\phi R_\nu(\pi) + D_\phi$ for each complete policy. For a fixed expert baseline, horizon, and positive action-sensitivity bound, reducing this combined quantity tightens the task-failure bound. This provides a conditional direction for improvement, not an ordering of actual success rates.

IV. Method

Section III motivates two directions: preserving information relevant to expert action selection and using new execution evidence to reduce remaining action error. Figure 1 organizes

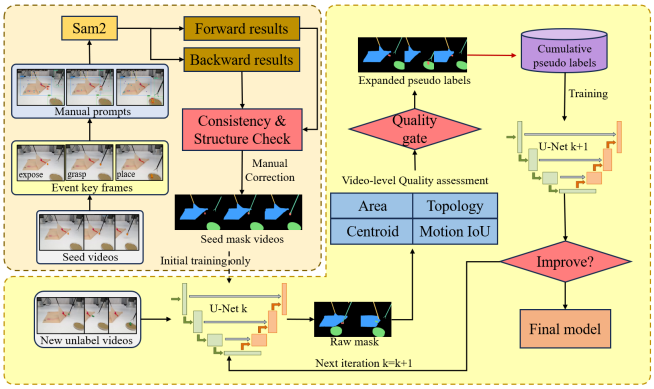


Fig. 2. Quality-aware iterative annotation. Reviewed bidirectional SAM2 masks train the initial segmenter. Subsequent models learn from cumulative, quality-screened pseudo-labels; seed videos are retained for reference rather than further training.

these directions into three components. First, quality-aware iterative annotation expands a small manually reviewed seed set into semantic supervision. Second, RGB observations, semantic maps, and supervised task-descriptor queries jointly support a base action-chunk policy. Third, a history-conditioned residual model uses updated observations to correct actions that have not yet been executed.

A. Quality-Aware Iterative Semantic Supervision

Task-descriptor labels are derived from semantic masks, so segmentation errors can compromise the supervision used to preserve task information. We combine reviewed seed annotations with video-level quality screening to limit error propagation as the labeled set expands (Fig. 2).

1) Event-Guided Seeds and Iterative Expansion: We select a small set of videos spanning the principal interaction and occlusion states, and annotate keyframes around events such as target exposure, contact, and placement. Bounding boxes and positive and negative point prompts guide bidirectional SAM2 propagation [23]; consistency checks, class-dependent structural processing, and manual correction produce the seed masks. The seeds train an initial U-Net segmenter, which predicts candidate masks on new unlabeled videos. Accepted masks accumulate across iterations, and each subsequent segmenter is reinitialized and trained on this cumulative pool without the original seed videos. The seed videos then serve as a fixed reference for monitoring segmentation accuracy and temporal anomalies. Iteration continues while anomaly rates improve without a material loss of segmentation accuracy.

2) Video-Level Quality Screening: The screening procedure distinguishes sustained task changes from transient segmentation anomalies. For each class, we extract trajectories of mask area, spatial extent, centroid, and topology, and fit robust piecewise trends over the video. Persistent changes can establish new segments, while short-lived departures provide anomaly evidence instead of penalizing every inter-frame change. Motion-compensated mask IoU supplies additional temporal evidence on pixels with forward–backward-consistent optical-flow correspondences. For class c at frame t, we collect these diagnostics as

$$\boldsymbol{\upsilon}_{t,c}^{\text{seg}} = (\upsilon_{t,c}^{\text{area}}, \upsilon_{t,c}^{\text{space}}, \upsilon_{t,c}^{\text{topo}}, \upsilon_{t,c}^{\text{motion}}). \tag{21}$$

Agreement across evidence groups identifies anomalous intervals; excessive changes introduced by structural processing also trigger exclusion. A frame enters subsequent segmentation training only when its relevant classes pass these checks. The manually reviewed reference set complements temporal screening by checking for persistent semantic errors that temporal consistency alone cannot detect. The same diagnostics also determine descriptor-specific supervision weights in Section IV-B. Quality assessment is performed offline; the final per-view segmenters are frozen and require only current images during policy execution.

B. Semantic Inputs and Task-Descriptor Queries

Semantic maps emphasize object-level spatial structure, while supervised queries encourage the joint representation to retain task-relevant attributes and relationships. RGB and proprioception remain available for decision cues not captured by the semantic classes. This design targets the descriptor-estimation term in (14) through the learned representation, rather than assuming that adding semantic channels improves coverage.

1) Joint Visual-Semantic Encoding: For view v, a frozen segmenter predicts class probabilities $p_t^v(c,\xi)$ from image I_t^v. We form a soft semantic image as

$$S_t^v(\xi) = \sum_{c \in \mathcal{C}_v} p_t^v(c,\xi) e_c^{\text{sem}}, \tag{22}$$

where $\mathcal{C}_v$ is the view-specific class set, ξ a pixel location, and $e_c^{\text{sem}} \in [0,1]^3$ a fixed class color. This probability-weighted color representation allows RGB and semantics to use the same three-channel visual encoder without argmax hardening. The encoded streams from all views are combined with proprioceptive and query tokens in a transformer encoder. An ACT decoder [6] reads this context to predict the base action chunk.

2) Supervised Descriptor Queries: We select K_{rel} descriptor groups $G^{(j)} \in \mathbb{R}^{d_j}$, such as visible area and relative position, which together form the task descriptor vector G. Fixed geometric operators applied to the offline masks generate pseudo-labels $\widetilde{G}_t^{(j)}$; the task-specific definitions are given in Appendix D-C. Each group has a learnable query whose encoded output $z_t^{(j)}$ supports a prediction head, $\widehat{G}_t^{(j)} = D_j(z_t^{(j)})$, and remains available to the action decoder. Supervision therefore encourages task information to be recoverable from the policy representation without prescribing an attention map.

A weight $\varpi_{n,j}^G \in [0,1]$ combines descriptor validity with quality evidence from the relevant semantic classes. Reliable labels receive full weight, uncertain labels are downweighted, and undefined or anomalous labels receive zero weight. In particular, a distance is undefined when either object is invisible, whereas a confirmed absence can provide a valid zero-visible-area label. Disabling a descriptor loss does not remove the

sample's action supervision. With descriptors normalized to fixed scales, the training objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{K_{\text{rel}}} \sum_{j} \frac{\sum_n \varpi_{n,j}^{G} \ell_G(\widehat{G}_n^{(j)} - \widetilde{G}_n^{(j)})}{\max(1, \sum_n \varpi_{n,j}^{G})},$$
$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{IL}} + \lambda_{\text{rel}} \mathcal{L}_{\text{rel}}. \tag{23}$$

Here, n indexes minibatch samples, ℓ_G is the component-averaged Smooth L1 loss, and $\mathcal{L}_{\text{IL}}$ is the standard ACT training objective. The segmenters remain frozen, and quality weights affect auxiliary supervision rather than online policy inputs. Appendices D-A and D-B specify label admission and its connection to descriptor-estimation error.

C. History-Conditioned Residual Action Feedback

Even with task-relevant information, actions generated from an earlier observation may not match current execution needs. Following Section III-D, we freeze the base policy and train a residual model to predict its remaining action error from updated observations, plan context, and execution history.

1) Plan-Conditioned History Encoding: The base policy generates an action chunk and query context at time τ_p. A feedback update at τ_o predicts corrections for future execution times $t \geq \tau_o$. Its input record e_k^{fb} combines current visual and proprioceptive features, cached query features and target base actions, and recent command–response information. Query features are cached between plan updates, while images are re-encoded at each feedback update; relative timing identifies the actions being corrected. A GRU encodes the most recent L_{hist} records:

$$h_i^{\text{gru}} = \text{GRU}_\omega(e_i^{\text{fb}}, h_{i-1}^{\text{gru}}). \tag{24}$$

The hidden state is initialized to zero at the start of each window, and its final value h_k^{gru} is used for prediction. This history allows the model to interpret the current observation in light of previous commands and their observed responses.

2) Residual Prediction and Supervision: For each of H_r future slots, the residual head predicts a bounded adjustment to the corresponding base action:

$$r_{k,j} = \boldsymbol{\beta} \odot \tanh([W_r h_k^{\text{gru}} + b_r]_j),$$
$$\widehat{u}_{k,j} = B_{t_{k,j}} + r_{k,j}. \tag{25}$$

Here, $t_{k,j}$ is the target execution time, $B_{t_{k,j}}$ is the base command for that slot, and $\boldsymbol{\beta}$ bounds the residual in each action dimension.

Training replays the frozen base policy on demonstration trajectories, matching the observation and target-slot timing used for execution. For sample n and slot j, the residual target is $E_{nj} = u_{nj}^{\text{E}} - B_{nj}$, the demonstrated command minus the base command at the same execution step. Over valid sample–slot pairs $\mathcal{V}$, we minimize

$$\mathcal{L}_{\text{res}} = \frac{1}{|\mathcal{V}|} \sum_{(n,j)\in\mathcal{V}} \left[\ell_r\left(\frac{r_{nj} - E_{nj}}{\boldsymbol{\sigma}}\right) + \frac{\lambda_r}{d_u} \| \frac{r_{nj}}{\boldsymbol{\sigma}} \|^2 \right], \tag{26}$$

where ℓ_r is component-averaged Smooth L1, $\boldsymbol{\sigma}$ contains training-set action scales, and division is elementwise. Future demonstrated commands provide labels only and are not feedback inputs. This robust training objective is a surrogate for error reduction; the squared-error criterion in Section III-D is evaluated separately on held-out data. Replay learns corrections on demonstration histories, while their effect on subsequent execution is evaluated in closed-loop trials.

3) Execution Alignment: Each residual is attached to its originating plan and target slot; corrections for replaced plans or elapsed slots are discarded. The executed command is $u_t = B_t + \Lambda_t r_t$, where Λ_t indicates plan and timing validity, so an unavailable correction leaves the base command unchanged. This alignment applies the correction to the action it was trained to modify; the timing convention is summarized in Appendix D-E.

V. Experimental Evaluation

This section evaluates the proposed method on a bimanual exposure and execution task. The task requires coordinated actions under deformable occlusion, where preceding interactions change the visibility and accessibility of the target. We first assess overall task performance across object categories, initial positions, and occluder colors, then examine annotation reliability, semantic representations, and history-conditioned correction through controlled ablations. Finally, we evaluate the framework on a surgical-phantom task to assess its applicability beyond the primary setting.

主要实验围绕四个问题展开：

1) 完整方法能否提高任务完成率，且收益是否在不同物体、初始位置和遮挡物颜色上保持一致？
2) 质量感知迭代能否在保持分割精度的同时减少异常，并降低人工标注负担？
3) 语义输入与任务描述量监督分别带来什么收益，感知质量如何影响这些收益？
4) 更新观测与历史能否减少执行错误，并促进自然偏差后的恢复？

A. Experimental Setup

1) Task and Platform: 平台采用两条 SO-ARM101 机械臂，右臂末端安装长板工具，通过固定的前视与侧视相机观察桌面操作区域。示范中的目标包括方块、纸团和螺丝螺母三类，遮挡物为同一颜色的可形变布料。任务开始时目标位于布料下方，双臂协同拨开布料，使目标暴露并与布料分离，再将目标推入指定区域，最后恢复布料在前视画面中的覆盖率。

以完整目标投影 $O_t^\star$、完整目标区域足迹 $R^\star$ 和布料投影 C_t 定义入区与复原条件：

$$p_{\text{in}}(t) = \mathbf{1}\{O_t^\star \subseteq R^\star\},$$
$$p_{\text{restore}}(t) = \mathbf{1}\{|C_t \cap \Omega| / |\Omega| \geq \theta_C\}, \tag{27}$$

其中 Ω 为固定前视相机的整幅图像域，θ_C 为布料覆盖阈值。任务成功要求目标完整入区且布料覆盖率达到阈值，两项条件须在规定时限内同时成立：

$$\Phi(s_{0:T}) = \mathbf{1}\{\exists\, t \leq T : p_{\text{in}}(t) = 1 \wedge p_{\text{restore}}(t) = 1\}. \tag{28}$$

（填写最大执行时间与布料覆盖阈值。）

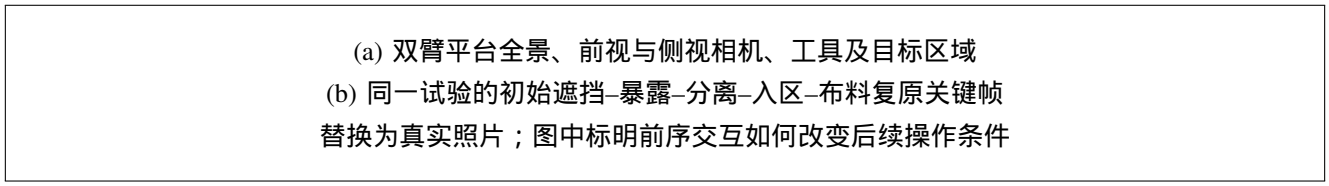


Fig. 3. Bimanual exposure and execution: the experimental platform and a complete task sequence.

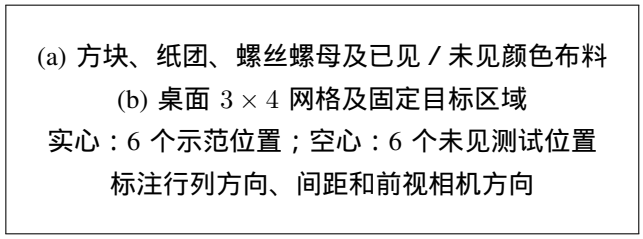


Fig. 4. Object categories, occluder colors, and initial positions. Filled and open markers indicate demonstration and unseen test positions, respectively.

TABLE I
Key training and execution settings.

项目	配置
数据划分 / 训练种子	待填
每个条件执行重复	待填
时限 / 布料覆盖阈值	待填
控制频率 / 部署设备	待填
动作块 / 重规划间隔	待填
残差间隔 / 预留延迟	待填

2) Demonstrations and Test Conditions: 数据集包含 90 条示范、53,723 个帧对齐样本，采集频率为 30 Hz。前视与侧视原始图像均为 640×360，本体状态和动作均为 10 维关节位置，对应每条机械臂的 5 个关节。

物体初始位置按 3×4 网格布置，示范覆盖其中六个位置，如图 4所示；基础测试覆盖三类已见物体与全部 12 个位置的组合，共 36 个条件。另使用未见颜色的遮挡布料，在相同物体与位置条件下检验外观变化的影响。各方法在每个条件下进行相同次数的独立重置试验，分别报告已见与未见位置的表现。

（确定未见布料颜色与重复次数；固定三类物体、12 个位置和各布料颜色的测试组合。）

3) Training Settings: 各方法在固定示范预算下使用相同的 RGB 观测、本体状态、动作接口和示范划分；语义方法增加由示范视频获得的语义监督。基础策略匹配训练更新次数与样本访问量，语义预训练和残差训练的额外资源单独报告。残差消融共用冻结的基础策略，模型选择与最终测试分开。表 I汇总关键执行设置，数据使用与评价细则见附录 D-D及E。

4) Evaluation Metrics: 主要指标为固定时限内的完整任务成功率，报告成功次数 / 总次数及不确定性区间。补充指标包括成功试验的完成时间、以全部试验为分母的累计成功时间曲线、阶段到达与条件完成率，以及主要失败类型。各方法采用一致的初始条件和计时规则，测试顺序在条件区组内随机安排。任务成功与阶段进展由独立视频审核判定。

B. Overall Task Performance

本节比较整体任务表现，并考察物体类别、初始位置和遮挡物颜色变化下的成功率。对照包括 ACT [6]、Diffusion Policy [8] 和完整方法；另预留 A2C2 [15] 作为在线动作块纠正对照、VIOLA [12] 作为对象中心表示对照。

（完成所选基线的合理调试后再纳入最终表格，报告适配改动与额外资源。ACT 上的 A2C2 适配不等于原版复现，本文 MLP 消融也不能直接标为 A2C2。）

各基线采用匹配的模型选择与调参预算。推理速度以部署设备上的端到端测量为准，不以名义更新频率代替实际频率。

1) Task Success and Completion Time: 表 II汇总三个已见物体类别、已见与未见位置、已见与未见遮挡物颜色的成功率，以及整体成功率和成功试验的平均完成时间。整体成功率以预先确定的完整测试池为分母，每次试验只计一次；分组列分别统计满足对应条件的试验。图 5进一步展示逐位置表现。

（先比较整体成功率与完成时间，再检查收益在三个物体类别上是否一致，以及未见位置和颜色下是否保持。结合全部试验的成功时间曲线，避免仅由成功样本均值判断效率。分别分析位置、颜色变化及两者同时变化的情况；结论限定为固定示范预算下的表现。）

2) Completion Time, Stage Progress, and Failures: 图 6比较完成时间与阶段到达情况，表 III汇总主要失败类型。二者用于区分暴露、目标执行与复原环节的改善，具体失败机制在 Discussion 中分析。

（比较完整方法相对各基线的成功率差值及区间，再结合时间曲线和阶段统计说明收益发生在哪个环节。若只在某类物体或初始条件下改善，直接限定结论范围。表 III只统计现象，具体原因结合视频与日志在 Discussion 分析；未完成的可选基线行在定稿时删除。）

TABLE II
Task success across object categories, initial positions, and occluder colors, with overall success and mean completion time. Success entries report count/total and percentage; time is computed over successful trials.

方法	类别成功率			位置成功率		遮挡物颜色成功率		整体表现	
	方块 已见	纸团 已见	螺丝螺母 已见	已见 6 位置	未见 6 位置	已见 颜色	未见 颜色	成功率	时间 (s)
ACT	–	–	–	–	–	–	–	–	–
Diffusion Policy	–	–	–	–	–	–	–	–	–
ACT + A2C2 适配	–	–	–	–	–	–	–	–	–
VIOLA 适配	–	–	–	–	–	–	–	–	–
Ours	–	–	–	–	–	–	–	–	–

行：方块、纸团、螺丝螺母；列：ACT、DP、Ours
每个面板为 3×4 位置成功率热图，统一色标，标出示范位置
每格标记成功次数 / 重复次数，按布料颜色分面；未测试组合留空

Fig. 5. Position-wise task success by object category and occluder color. Highlighted cells indicate demonstration positions; untested combinations remain blank.

(a) 截至时间 t 的完整成功比例曲线
(b,c) 暴露、分离、入区、复原的累计到达率 / 条件完成率
按方法绘制，包含误差区间和分母；不将失败试验从时间曲线中删除

Fig. 6. Completion-time and stage-level comparisons. Placeholder for measured curves and stage statistics.

TABLE III
Primary failure categories for the everyday task. Report counts and fractions of all trials for each method.

方法	主要失败类型	次数 / 全部试验
待填	未完成暴露	–
待填	无效接触 / 拨空气	–
待填	目标移出可操作范围	–
待填	入区不完整	–
待填	复原不足	–
待填	其他 / 原因未确定	–

(a) 累积标注量–精度 / 异常率曲线
(b) 同一视频的质量轨迹与真实事件标记
(c) 异常帧的原始预测、处理结果和人工参考
保留真实暴露 / 接触变化作为对照

Fig. 7. Annotation expansion and temporal quality diagnostics. Placeholder for measured curves and reviewed mask examples.

C. Quality-Aware Semantic Annotation

本节检验质量感知迭代能否在保持分割精度的同时减少异常，并降低获得可靠标签的人工成本。按完整视频划分标注开发数据与独立人工审核数据。比较种子模型、无质量筛选的迭代和质量筛选迭代；两条迭代路径使用相同候选视频池及匹配训练预算，并记录实际保留的数据量。独立人工审核视频包含普通帧以及暴露、接触和入区等关键事件，两类样本分层报告，评价不以迭代自身的质量分数代替人工参考。

表 IV报告原始输出的 mIoU、异常率、可用标签比例与人工投入。异常帧由预先定义的漏分、错分和非真实结构突变规则审核；时间一致性使用有效光流对应区域的运动补偿 IoU。原始输出与后处理结果分开统计，以区分模型本身的改善和筛选 / 修补的作用。按类别补充目标和工具的 IoU 及异常计数，跨视频指标以视频为单位汇总。人工成本计入关键帧标注、检查与修正时间，并结合达到统一审核标准的可用标签数量报告。在相同视频子集上记录人工逐帧标注 / 修正流程的耗时，作为人工成本参照。

（先比较无筛选与有筛选迭代，再分析每轮精度、异常率、保留率及人工成本的变化。不能把排除了更多帧等同于模型更稳定；分别说明原始预测与后处理的贡献。如精度基本持平而异常下降，应以稳定性改善为结论，不写成空间精度显著提升。填写人工耗时记录与对照子集；没有实际耗时时仅报告异常和复核需求，暂不声称节省人工时间。）

（核对用于该实验的分割训练记录。现有数据集的前视标注继承旧版本，侧视来自另一标注流程，不能将已有标签直接记为最新迭代方法的产物；必要时在相同原始视频上补做匹配实验。分别报告前视与侧视结果，人工成本同时计入种子标注和后续复核。）

TABLE IV
Annotation quality and human effort on independently reviewed everyday-task videos. Iteration index k is not fixed in advance.

标注 / 分割版本	标注视频数	raw mIoU	原始异常率	运动 IoU	保留标签率	人工分钟 / 视频
种子模型	–	–	–	–	–	–
无质量筛选迭代	–	–	–	–	–	–
质量筛选迭代 1	–	–	–	–	–	–
质量筛选迭代 k	–	–	–	–	–	–

D. Semantic Representation Ablations

本节在不启用残差反馈的条件下，区分语义输入、查询容量和描述量监督的贡献。各组在三类物体和 12 个位置上采用相同的测试条件，主要消融沿用已见颜色的布料。正文报告分类别、已见 / 未见位置成功率以及阶段完成率和失败类型，完整逐条件结果放补充材料。表 V中 S0–S3 逐步加入语义与查询；S2 和 S3 保持查询数量及结构一致，只改变描述量辅助监督。S3 与 S4 使用相同标签和分割器，仅改变质量权重。S4 停用质量筛选与降权，但保留相同的几何有效性标记，不对无定义距离进行监督。这些配置分别训练基础策略，并保持动作监督、视角及预算一致。

S5 固定 S3 的策略权重，仅在推理时替换为验证集上预先选定的较低质量分割器，保持类别定义、输入尺寸与语义编码一致，以评价在线感知质量的影响。分割器更换造成的延迟变化单独记录。此组不同时改变训练标签；不同质量标签训练出的策略属于另一种对照，不能混合解释。

图 8按物体与位置汇总任务结果，并定位阶段收益与失败类型的变化。进一步在独立审核数据上评价伪标签误差和部署模式下的描述量读出误差，按面积、距离和方向分别报告，并保留有效标签比例。补充的专家动作重构探针用于考察描述量是否保留动作选择所需的信息，其划分与对照见附录 E-E。该分析检验第 III-C节提出的任务信息保留路径。

（S1–S0 检验语义输入；S3–S2 检验描述量监督是否超出查询容量本身的作用；S3–S4 检验质量权重；S5–S3 检验感知退化影响。先检查分类别与已见 / 未见位置的趋势，再结合阶段、失败类型和描述量误差解释收益。若低质量感知削弱收益，报告其具体影响环节。读出 / 重构结果用于解释信息保留，不直接估计理论覆盖常数。）

TABLE V
Representation ablations without residual feedback. UQ: queries without descriptor supervision; SQ: supervised task-descriptor queries.

ID	输入 / 查询	质量权重	推理分割	策略重训	成功率	端到端 p95(ms)
S0	RGB	不适用	无	是	–	–
S1	RGB + 语义	不适用	标准	是	–	–
S2	RGB + 语义 + UQ	不适用	标准	是	–	–
S3	RGB + 语义 + SQ	启用	标准	是	–	–
S4	RGB + 语义 + SQ	停用	标准	是	–	–
S5	同 S3	同 S3	较低质量	否	–	–

(a) S0–S5 三个类别及已见 / 未见位置的成功率
(b) 阶段累计到达率与条件完成率；(c) 主要失败类型占全部试验的比例
(d) 伪标签与模型读出的分类描述量误差及有效标签比例
标明样本量和区间；专家动作重构探针结果放附录

Fig. 8. Representation ablations: task success, stage progress, failure categories, and descriptor accuracy.

E. History-Conditioned Residual Feedback

本节固定同一个 RGB + 语义 + QToken 基础 checkpoint，检验语义任务上下文与历史能否减少执行错误，并帮助策略从自然偏差中恢复。表 VI采用 MLP / GRU 与是否读取语义 + QToken 信息的交叉设计；语义特征和 QToken 特征在本消融中作为组合加入，不分别归因。

1) Matched Inputs and Residual Models: 全部残差组读取最新 RGB、本体状态、当前基础计划及目标槽位信息。不读取语义组合的组也不读取融合语义或 QToken 后的 Transformer 特征；基础动作仍来自共同的语义增强策略。语义组合组额外读取语义特征与缓存的 QToken 特征，分别记录其对应的观测时刻，不将缓存查询视为最新观测。

（确认残差语义特征来自本次分割还是缓存；固定这项实现后再训练所有组合组，报告额外分割带来的实际时延。）

MLP 使用当前记录，GRU 使用包含相同当前记录的历史窗口。各组分别训练残差头，匹配目标动作、更新间隔、输出长度、限幅与训练划分；历史携带的速度或上一指令等信息在各组输入定义中明确。主要组复核容量接近的当前信息模型，避免仅将参数增加解释为历史收益。

F0 与 S3 使用同一基础策略，F4 为完整方法。另在相同冻结基础策略上接入 A2C2 适配模块，作为外部修正方法对照，匹配可用输入与训练预算，并记录其原有逐步修正方式与本文执行方式的实际时延差异。另以 RGB ACT + RGB 历史残差作为连接对照，与 S0、S3 和 F4 构成表示增强与反馈的整体比较。为比较执行方式，在同一基础模型上保留更频繁重规划与 temporal ensemble 对照，并记录实际算力和更新频率，而不默认其与残差具有相同计算成本。

2) Error Reduction and Natural Recovery: 各组沿用表示消融的物体与初始位置条件，不施加人工扰动。除分类别、已见 / 未见位置成功率外，比较阶段完成率和主要失败类型，并从完整执行视频中记录布料缠阻、无效接触、目标输送偏差和执行停滞四类事件。这些事件可出现在成功或失败试验中；局部错误后的恢复与最终任务成功分别记录。

错误发生率以出现至少一次错误的试验数除以全部试验数；自然恢复率以预定窗口内恢复的事件数除以全部已识别事件数，并报告出错试验的最终成功率和恢复耗时。无错误事件时恢复率记为不适用，不记为零。表 VII按错误类型统计上述结果。事件起止、恢复条件和重复计数规则见附录 E-D。不同策略经历的错误状态可能不同，因此自然恢复统计与相同基础策略上的总体成功率、错误发生率结合解释。

3) Correction Accuracy and Runtime: 离线纠偏评价使用相同回放样本、专家动作目标、执行槽位和动作尺度，报告基础动作与修正动作的 MSE 及式 (43)中的对齐收益和修正能量。表 VIII记录端到端延迟、观测年龄、实际更新频率和过期修正比例。候选修正与实际执行指令分别记录；控制组若保留反馈计算但屏蔽输出，可以进一步检查计算调度变化是否影响基础策略。

（F2–F1 与 F4–F3 检验语义 + QToken 组合信息；F3–F1 与 F4–F2 检验历史。先看同槽动作误差，再看恢复及最终成功，区分预测改善、修正被及时采用和闭环收益。同时检查错误发生率：若错误更少，即使恢复事件减少也可能体现反馈收益。与同基础策略的 A2C2 比较修正效果和端到端延迟；记录 p50 / p95、观测年龄与过期比例。若某组更慢或出现有害修正，结合延迟和修正能量报告，不以更平滑的轨迹代替有效纠偏。失败机制在 Discussion 结合独立证据解释。）

TABLE VI
Residual ablations and a correction-method comparison with one frozen semantic-query base policy. Error incidence and recovery are measured from naturally occurring execution errors.

ID	残差模型	语义 + QToken	历史	成功率	错误发生率	自然恢复率	动作 MSE
F0	无残差	不适用	不适用	–	–	–	–
F1	MLP	否	否	–	–	–	–
F2	MLP	是	否	–	–	–	–
F3	GRU	否	是	–	–	–	–
F4	GRU	是	是	–	–	–	–
对照	A2C2 适配	匹配	原方法	–	–	–	–

TABLE VII
Natural-error and recovery diagnostics. Instantiate each method separately. Final success is measured over trials containing the specified error; recovery time is conditional on event recovery.

方法	错误类型	出错试验 / 总数	恢复 / 事件数	出错试验最终成功率	恢复时间 (s)
待填	布料缠阻	–	–	–	–
待填	无效接触	–	–	–	–
待填	输送偏差	–	–	–	–
待填	执行停滞	–	–	–	–
待填	去重合计	–	–	–	–

TABLE VIII
Correction accuracy and runtime. Offline errors use matched evaluation samples; runtime is measured on the deployment device.

修正方法	对齐收益	修正能量	延迟 p50 / p95(ms)	观测年龄 (ms)	更新频率 (Hz)	过期比例
F1	–	–	–	–	–	–
F2	–	–	–	–	–	–
F3	–	–	–	–	–	–
F4	–	–	–	–	–	–
A2C2 适配	–	–	–	–	–	–

(a) F0–F4 分类别、已见 / 未见位置成功率及阶段完成率
(b) 失败类型、出错试验最终成功率及自然恢复结果
(c) 自然错误–修正–恢复 / 未恢复视频序列与同步指令轨迹
标明错误起点、恢复时刻及计划 / 反馈更新时间；同时展示减少错误与恢复失败案例

Fig. 9. Error reduction and natural recovery with residual feedback. Placeholder for task results and synchronized visual and command traces.

F. Cross-Task Applicability

最后检验相同框架能否支持日常推动以外的操作。本节在手术模拟任务数据上分别训练各方法，保留语义监督、任务描述量查询和历史残差的整体设计，评价对象是方法的跨任务适用性，而非同一策略的零样本迁移。

1) Surgical-Phantom Task and Adaptation: 该任务使用组织模拟材料遮挡目标，由双臂协同完成牵拉暴露、夹取和向指定区域转移。相较日常任务，接触方式由推动转为夹持，且暴露动作须与后续抓取协调。以目标完整进入指定区域并完成释放作为拟定终止条件，过程评价记录暴露、稳定夹持和放置三个里程碑。

（确认最终成功条件是否还要求组织复原，填写时限、持续确认时间、机器人与动作接口、物体和初始条件、数据规模、独立测试重复以及安全停止规则；图表仅为待完成实验的预留。）

各方法使用相同的新任务示范和评价条件，模型选择与测试分开。需要适配的部分包括机器人动作接口、语义类别以及任务描述量的实例定义；训练流程与模型设计的共用部分和实际改动分别报告。

（列出实际保留与调整的设计、分割标注人工成本和新增训练预算；若更换描述量、相机布置或控制频率，明确其原因，不将任务相关适配表述为无需修改。）

(a) 手术模拟平台、器械、相机与材料
(b) 遮挡–牵拉暴露–稳定夹持–转移放置
使用同一次试验的关键帧，标明成功终止状态

Fig. 10. Surgical-phantom exposure and execution as an additional application. Placeholder for the platform and task sequence.

2) Overall Comparison and Key Ablations: 表 IX比较 ACT、DP 与完整方法，并保留语义输入、语义 + QToken 无残差、语义 + QToken 配合 MLP 残差三项关键对照。前两项检查表示收益是否延续，MLP 与 GRU 的比较检查历史是否仍有帮助，不在此重复日常任务的全部消融。残差组共用在手术模拟数据上训练的语义查询基础策略。

（先比较完整方法与 ACT、DP，再检查表示和历史残差的收益是否与日常任务一致。结合各阶段结果说明提升是否发生在暴露、夹持或放置；若趋势不同，分析任务接触要求与感知条件的差异。若放置已是全部终止条件，合并表中放置率与完整成功率以避免重复。结论限于两个应用中实际测试的条件，不表述为未训练策略的跨任务泛化或临床有效性。）

TABLE IX
Cross-task applicability on surgical-phantom execution. Models are trained on this task's demonstrations. Stage entries are cumulative attainment rates; time is conditional on full success.

方法	完整成功率	时间 (s)	暴露率	夹持率	放置率
ACT	–	–	–	–	–
Diffusion Policy	–	–	–	–	–
ACT + 语义	–	–	–	–	–
ACT + 语义 + QToken	–	–	–	–	–
ACT + 语义 + QToken + MLP 残差	–	–	–	–	–
Ours (GRU 残差)	–	–	–	–	–

VI. Discussion and Conclusion

A. Failure Cases and Limitations

本节结合表 III的失败分布和同步视频、指令、感知日志，分析完整方法尚未解决的问题。案例选择覆盖不同失败类型，也包含残差未能恢复及修正后进展倒退的情况，不以个别成功恢复片段替代整体统计。

1) Perception and Task Information: 质量筛选针对视频中的异常变化，不能排除时间上稳定的系统性误分。所选描述量刻画对象属性与空间关系，但不一定足以解释接触状态或专家的具体动作选择；语义分割准确与描述量读出准确也不意味着策略已经正确利用这些信息。

(结合图 11(a,b) 区分分割错误、描述量估计错误和执行决策错误。说明是否存在描述量准确仍失败的情况，并联系理论中的专家选择近似误差解释未被当前信息覆盖的动作选择因素。没有探针或干预证据时，将原因表述为待验证解释，而非确定归因。)

2) Residual Prediction and Execution: 历史提供新的动作–响应证据，但有限数据、修正幅度和执行延迟可能限制其作用。依据第 III-D节，实际收益取决于可预测误差与残差实现误差的关系，而不是仅由是否使用 GRU 决定。

(结合图 11(c) 和表 VII区分预测不准、时效不足、修正过大和超出可恢复范围的案例；比较 MLP 与 GRU 在哪些条件下出现不同结果。修正之后发生失败仅说明时间关联，若无匹配对照，不直接声称由残差造成。)

3) Scope of the Evidence: 本文的任务表现界使用总体模仿误差，并依赖覆盖、动作敏感性和专家选择条件。独立轨迹上的读出和动作误差用于检验相应机制，不能将有限训练损失直接代入这些界，也不能据更紧的上界确定实际成功率排序。两个任务上的分别训练验证方法的适用性，不等同于零样本跨任务迁移；手术模拟材料上的实验亦不构成临床有效性证据。

(根据真实划分与训练重复说明结论覆盖哪些物体、位置和平台。联系理论 B–D 总结哪些机制获得实验支持、哪些条件尚未得到直接测量；概括新增任务的描述量设计、语义标注与训练成本。若未做多预算实验，不声明样本效率优势；QToken 没有独立停止头时，不将任务成功等同于自主停止能力。)

B. Conclusion

(用一段总结问题、三个方法模块及两个实机任务支持的核心结论。填写实际成功率、稳定性或恢复收益，只保留与匹配对照相符的表述；最后指出一个由失败分析直接支持的主要局限，不预先声明显著提升、通用性或收敛保证。)

(a) 持续分割错误：RGB、语义预测、人工审核
(b) 描述量准确但执行失败：任务关系与接触状态
(c) 未恢复或进展倒退：基础计划、残差、实际运动与时间戳
填写两个任务的代表性案例；只标注日志与视频支持的事实

Fig. 11. Representative failure cases and supporting evidence. Placeholder for perception, task-information, and residual-execution limitations.

Appendix A

A. Proof of the Policy Performance-Difference Identity

We prove (8) by summing expected changes in the reference value. The evaluation-domain restriction in Section III-A and Assumption 1 give the terminal value

$$V_T^{\pi_0}(\mathcal{H}_T) = 1 - \mathcal{V}(I_{0:T}^{1:V}) = 1 - \Psi. \tag{29}$$

For $t < T$, taking action $\pi(\mathcal{H}_t)$ and then following π_0 gives the failure probability $V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})$ conditional on the next history. Averaging over next histories under Assumption 2 gives

$$Q_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = \mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]. \tag{30}$$

Here, π determines the next-history distribution, while $V_{t+1}^{\pi_0}$ accounts for subsequent execution under π_0.

Combining (7) and (30) and taking expectations gives

$$\begin{aligned}&\mathbb{E}_\pi[A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))]\\ &= \mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)].\end{aligned} \tag{31a}$$

By the tower property, averaging the conditional next-step value over current histories gives its unconditional expectation:

$$\begin{aligned}&\mathbb{E}_\pi\left[\mathbb{E}_\pi\left[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1}) \mid \mathcal{H}_t\right]\right]\\ &= \mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})].\end{aligned} \tag{31b}$$

Substituting (31b) into (31a) and summing over time leaves only the terminal and initial terms:

$$\begin{aligned}&\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t))\\ &= \sum_{t=0}^{T-1}\left\{\mathbb{E}_\pi[V_{t+1}^{\pi_0}(\mathcal{H}_{t+1})] - \mathbb{E}_\pi[V_t^{\pi_0}(\mathcal{H}_t)]\right\}\\ &= \mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] - \mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)].\end{aligned} \tag{31c}$$

At time T, no action remains. By (29), the terminal term is the failure probability under π:

$$\mathbb{E}_\pi[V_T^{\pi_0}(\mathcal{H}_T)] = \mathbb{E}_\pi[1 - \Psi] = F(\pi). \tag{31d}$$

The policies share the initial-history distribution, so the initial term equals the failure probability of following π_0 throughout:

$$\begin{aligned}\mathbb{E}_\pi[V_0^{\pi_0}(\mathcal{H}_0)] &= \mathbb{E}_{\pi_0}[V_0^{\pi_0}(\mathcal{H}_0)]\\ &= F(\pi_0).\end{aligned} \tag{31e}$$

Finally, substituting (31d) and (31e) into (31c) gives

$$\mathbb{E}_\pi \sum_{t=0}^{T-1} A_t^{\pi_0}(\mathcal{H}_t, \pi(\mathcal{H}_t)) = F(\pi) - F(\pi_0). \tag{31f}$$

This establishes (8).

B. Proof of the Task-Failure Bound

We prove (9) by first bounding task consequences through action deviations and then transferring the resulting deployment error to the demonstration distribution.

Set $\pi_0 = \pi_{\mathrm{E}}$ and let $e(h) = \|\pi(h) - \pi_{\mathrm{E}}(h)\|$. Applying Assumption 4 in (8) bounds each step's increase in failure probability by its action deviation:

$$F(\pi) - F(\pi_{\mathrm{E}}) \le L\,\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right]. \tag{32a}$$

Since $\bar{\mu}_\pi$ samples control steps uniformly, the time sum equals T times the deployment average:

$$\mathbb{E}_\pi\left[\sum_{t=0}^{T-1} e(\mathcal{H}_t)\right] = T\,\mathbb{E}_{(t,h)\sim\bar{\mu}_\pi} e(h). \tag{32b}$$

By the Cauchy–Schwarz inequality, the mean action deviation is bounded by its root mean square:

$$\mathbb{E}_{\bar{\mu}_\pi} e(h) \le \sqrt{\mathbb{E}_{\bar{\mu}_\pi} e(h)^2}. \tag{32c}$$

Assumption 3 also bounds expectations of nonnegative measurable functions, by simple-function approximation and monotone convergence. Applying it to $e(h)^2$ and using (3) gives

$$\mathbb{E}_{\bar{\mu}_\pi} e(h)^2 \le C\,\mathbb{E}_\nu e(h)^2 = C R_\nu(\pi). \tag{32d}$$

Combining (32a)–(32d) establishes (9).

C. Proof of the Representation-Level Transfer Bound

We prove (10) by separating conditional error discrepancy from distributional coverage, then apply action sensitivity to obtain (11).

Let $\mu = \bar{\mu}_\pi$, with the remaining notation and the integrability and measurable conditional-mean conditions as in Section III-B3. Representation coverage ensures that ν_Z-null sets are also μ_Z-null sets, so the demonstration-side conditional mean is well defined for the deployment comparison.

First, conditioning on the representation gives

$$\mathbb{E}_\mu e(h)^2 = \mathbb{E}_{\mu_Z} m_\mu(Z). \tag{33a}$$

At the same representation, deployment and demonstration conditional errors can differ. By the definition of D_ϕ,

$$\begin{aligned}\mathbb{E}_{\mu_Z} m_\mu(Z) &\le \mathbb{E}_{\mu_Z} m_\nu(Z) + \mathbb{E}_{\mu_Z}|m_\mu(Z) - m_\nu(Z)|\\ &= \mathbb{E}_{\mu_Z} m_\nu(Z) + D_\phi.\end{aligned} \tag{33b}$$

Finally, applying representation coverage to the nonnegative function m_ν and taking the total expectation give

$$\mathbb{E}_{\mu_Z} m_\nu(Z) \le C_\phi \mathbb{E}_{\nu_Z} m_\nu(Z) = C_\phi R_\nu(\pi). \tag{33c}$$

Combining (33a)–(33c) establishes (10). Under Assumption 4, substituting this result into the sensitivity, time-average, and Cauchy–Schwarz steps (32a)–(32c) yields (11).

Appendix B
Proofs for Task-Relevant Information Preservation

A. Proof of the Expert-Action Reconstruction Bound

We prove (14) under the common encoder and readouts of Section III-C, with $\mu = \bar{\mu}_\pi$. Define the common represented expert rule $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$. The triangle inequality and Assumption 5 give, for $\rho \in \{\mu, \nu\}$,

$$\begin{aligned}\mathbb{E}_\rho\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| &\le \mathbb{E}_\rho\|\pi_{\mathrm{E}} - g_{\mathrm{E}}(G,\chi)\| \\ &\quad + \mathbb{E}_\rho\|g_{\mathrm{E}}(G,\chi) - g_{\mathrm{E}}(\widehat{G},\chi)\| \\ &\le \epsilon_{\mathrm{sel},\rho} + K_{\mathrm{E}}\delta_{G,\rho} = b_\rho.\end{aligned} \tag{34}$$

B. Proof of the Conditional-Error Transfer Bound

We prove Proposition 1 by comparing actual imitation error with a common function of Z, then transferring the demonstration-side reconstruction error through representation coverage. Use $\pi(h) = p(Z)$ and $\widehat{a}_{\mathrm{E}}(Z) = g_{\mathrm{E}}(\widehat{G}, \chi)$ from Section III-C. Since $\pi(h) = p(\phi(t,h))$, the represented squared error $\ell_Z(z) = \|p(z) - \widehat{a}_{\mathrm{E}}(z)\|^2$ is the same function of z under both distributions. All three actions belong to $\mathcal{A}$, so the reverse triangle inequality yields

$$\begin{aligned}|e(h)^2 - \ell_Z(Z)| &\le \big(\|\pi - \pi_{\mathrm{E}}\| + \|\pi - \widehat{a}_{\mathrm{E}}\|\big)\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \\ &\le 2B_{\mathcal{A}}\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\|.\end{aligned} \tag{35}$$

Conditioning separately under each distribution gives

$$|m_\rho(z) - \ell_Z(z)| \le 2B_{\mathcal{A}}\mathbb{E}_\rho[\|\pi_{\mathrm{E}} - \widehat{a}_{\mathrm{E}}\| \mid Z = z]. \tag{36}$$

Insert the common function ℓ_Z between m_μ and m_ν and apply representation coverage to the nonnegative demonstration-side conditional error:

$$\begin{aligned}D_\phi &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + \mathbb{E}_{\mu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le \mathbb{E}_{\mu_Z}|m_\mu(Z) - \ell_Z(Z)| \\ &\quad + C_\phi\mathbb{E}_{\nu_Z}|m_\nu(Z) - \ell_Z(Z)| \\ &\le 2B_{\mathcal{A}}(b_\mu + C_\phi b_\nu).\end{aligned} \tag{37}$$

Representation coverage ensures that the demonstration-side conditional inequality also holds μ_Z-almost everywhere. This proves (15); substitution into (11) establishes (16).

Appendix C
Proofs for Information-Conditioned Residual Corrections

All proofs fix the base policy, expert targets, and distribution ν from Section III-D.

A. Proof of the Optimal Residual Predictor

For any square-integrable $\mathcal{F}$-measurable vector v, the tower property gives

$$\begin{aligned}\mathbb{E}_\nu\langle E - r^*_{\mathcal{F}}, v\rangle &= \mathbb{E}_\nu\langle\mathbb{E}_\nu[E - r^*_{\mathcal{F}} \mid \mathcal{F}], v\rangle \\ &= 0.\end{aligned} \tag{38}$$

For any square-integrable $\mathcal{F}$-measurable predictor r, write $E - r = (E - r^*_{\mathcal{F}}) + (r^*_{\mathcal{F}} - r)$. Expanding the square and applying (38) yields

$$\mathbb{E}_\nu\|E - r\|^2 = R^*_\nu(\mathcal{F}) + \mathbb{E}_\nu\|r - r^*_{\mathcal{F}}\|^2. \tag{39}$$

The second term is nonnegative and vanishes at $r = r^*_{\mathcal{F}}$, proving (18).

B. Proof of the Information-Gain Identity

For $\mathcal{F}_0 \subseteq \mathcal{F}_1$, the tower property gives $\mathbb{E}_\nu[r^*_{\mathcal{F}_1} \mid \mathcal{F}_0] = r^*_{\mathcal{F}_0}$. Since $r^*_{\mathcal{F}_0}$ is also $\mathcal{F}_1$-measurable, applying (39) with information $\mathcal{F}_1$ and predictor $r^*_{\mathcal{F}_0}$ gives

$$R^*_\nu(\mathcal{F}_0) = R^*_\nu(\mathcal{F}_1) + \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2. \tag{40}$$

Rearranging proves (19). The gain is strictly positive exactly when the two conditional means differ on a set of positive ν-probability.

C. Proof of the Realized-Correction Criterion

Setting $r = 0$ in (39) gives

$$R_\nu(\pi_{\mathrm{b}}) = R^*_\nu(\mathcal{F}) + \mathbb{E}_\nu\|r^*_{\mathcal{F}}\|^2. \tag{41}$$

Setting $r = r_{\mathcal{F}}$ in the same identity gives $R_\nu(\pi_{\mathcal{F}}) = R^*_\nu(\mathcal{F}) + \epsilon_{\mathrm{res}}(\mathcal{F})$. Combining these expressions proves (20). For two learned corrections under nested information, subtraction further gives

$$\begin{aligned}R_\nu(\pi_{\mathcal{F}_0}) - R_\nu(\pi_{\mathcal{F}_1}) &= \mathbb{E}_\nu\|r^*_{\mathcal{F}_1} - r^*_{\mathcal{F}_0}\|^2 \\ &\quad + \epsilon_{\mathrm{res}}(\mathcal{F}_0) - \epsilon_{\mathrm{res}}(\mathcal{F}_1).\end{aligned} \tag{42}$$

Thus, the predictive gain must exceed any additional realization error for richer information to improve the learned policy.

Equivalently, direct expansion gives an error diagnostic without estimating the ideal predictor:

$$R_\nu(\pi_{\mathrm{b}}) - R_\nu(\pi_{\mathcal{F}}) = 2\mathbb{E}_\nu\langle E, r_{\mathcal{F}}\rangle - \mathbb{E}_\nu\|r_{\mathcal{F}}\|^2. \tag{43}$$

Here $r_{\mathcal{F}}$ includes output processing and is zero when no correction is applied; it must remain measurable with respect to the available information. For admissible actions, apply (11) to each complete policy using its own deployment distribution and representation. A reduction in R_ν tightens that bound when the other terms are held fixed, or more generally when $C_\phi R_\nu + D_\phi$ decreases with a common expert baseline, horizon, and positive sensitivity bound. The identities concern population error under ν; replacing it with training loss requires a separate generalization argument.

Appendix D
Method Details

A. Descriptor Labels and Quality Weights

Descriptor supervision uses the quality evidence relevant to its geometric definition rather than a single frame-wide score. Visible area depends primarily on target extent and topology; relative position also depends on the localization of both objects, and a tool-endpoint vector is sensitive to tool shape. For the classes $\mathcal{C}_j$ used by descriptor j, let $\eta^{\text{seg}}_{t,c}$ indicate that class c passes anomaly screening, and let $m^G_{t,j}$ indicate that the descriptor is defined. The supervision weight is

$$\varpi^G_{t,j} = m^G_{t,j} \prod_{c \in \mathcal{C}_j} \eta^{\text{seg}}_{t,c}\, g_j(\upsilon^G_{t,j}), \tag{44}$$

where $\upsilon^G_{t,j}$ aggregates the relevant evidence in (21), and g_j is a nonincreasing weight between zero and one. Reviewed descriptor labels determine the reliable, downweighted, and rejected ranges; these weights are not probabilities of correctness. Without sufficient evidence to calibrate a graded weight, $g_j \equiv 1$ gives validity and anomaly admission alone. The groupwise normalization in (23) primarily changes the relative contributions of samples within a descriptor group.

An empty mask is a valid zero-area label only when absence is confirmed by review or supporting temporal evidence. Undefined distances and directions are excluded from the auxiliary loss rather than labeled as zero. Missing optical-flow correspondences provide no motion evidence, whereas excessive structural repair excludes a frame from segmentation training. Whole-video diagnostics may use later frames offline; they are not supplied to the deployed policy.

B. Pseudo-Label and Descriptor-Readout Errors

For either evaluation distribution ρ used in Section III-C, inserting the descriptor pseudo-label $\widetilde{G}$ gives

$$\delta_{G,\rho} \le \mathbb{E}_\rho \|\widehat{G} - \widetilde{G}\| + \mathbb{E}_\rho \|\widetilde{G} - G\|. \tag{45}$$

The terms separate descriptor fitting from pseudo-label error, motivating quality screening in addition to query supervision. When measured only on admitted labels, this comparison concerns the admitted subset; the admission rate is reported alongside the error.

C. Task-Descriptor Instance

Our task instance uses front-view masks for three descriptor groups, irrespective of whether the policy receives one or two camera views. They describe target visibility, target–region distance, and the vector from a tool endpoint to the target. For image width W and height H_I, let M_O, M_R, M_T be the object, region, and tool masks, with centroids c_O, c_R, c_T when nonempty. The labels are

$$\begin{aligned} \widetilde{G}^{(1)} &= |M_O|/(WH_I), \\ \widetilde{G}^{(2)} &= \|c_O - c_R\| / \sqrt{(W-1)^2 + (H_I-1)^2}, \\ \widetilde{G}^{(3)} &= \text{clip}(\bar{c}_O - \bar{e}_{T,L}, -1, 1). \end{aligned} \tag{46}$$

Here, $\bar{c} = (c_x/(W-1), c_y/(H_I-1))$, with rightward and downward positive directions. The endpoint $\bar{e}_{T,L}$ is the leftmost of the two extremes along the tool mask's principal axis in normalized coordinates. This geometric convention need not identify the physical contact tip. Target–region centroid distance is an auxiliary descriptor, not the criterion for complete target containment.

D. Evaluation Scope

Seed videos used by the initial segmenter provide a development reference, not an independent final test. Cross-video evaluation requires held-out videos excluded from annotation-model fitting and selection, with the split maintained through policy and residual training. For existing checkpoints, the documented residual train–validation split establishes only residual-head validation; it does not establish that upstream models were trained on disjoint episodes. Descriptor evaluation and residual-cache generation use the base policy in deployment mode. For ACT, this sets the latent input to zero rather than conditioning on the demonstrated action posterior [6].

E. Residual Targets and Execution Alignment

Residual targets use the same demonstrated command stream and action scale as the base policy; measured joint states remain observations. Training normalizes errors by per-dimension action scales, while residuals are added to the base command in matching command units. On the discrete control grid, a feedback observation at step n_k with reserved delay d_{fb} targets

$$t_{k,j} = n_k + d_{\text{fb}} + j, \qquad j = 0, \dots, H_r - 1. \tag{47}$$

Online timestamps determine whether each target slot is still available; a late correction is not shifted onto a different action. Plan replacement invalidates residuals attached to the previous plan. Timing evaluation measures observation-to-execution latency, including acquisition, encoding, and scheduling, rather than network inference alone.

（定稿前须完成统一数据划分及新版描述量质量权重的实现核对；已有结果不能自动作为新版方法的验证。待核对事项单列于 method_implementation_notes.md。）

Appendix E
Evaluation Details

A. Data and Input Details

前视标注包含遮挡物、目标、目标区域、工具和左右机械臂六个前景类别；侧视标注包含前四类，各视角另含背景类。

（按最终 checkpoint 补充动作监督来源、归一化与图像预处理。当前数据集元信息将 90 条示范全部列入训练范围，按日志核对实际划分与模型选择依据；如设置验证集，应按完整轨迹划分，避免相邻帧泄漏。采集频率与实际控制、推理频率分别报告。）

TABLE X
Evaluation milestones for everyday bimanual exposure and execution.

里程碑	评价含义
暴露	目标由遮挡转为可辨识可见，达到预设可见程度。
分离	目标与布料的可见边界不再邻接，且目标具有足够可见性以支持判断。
入区	完整目标位于独立参考轮廓定义的目标区域内。
复原	布料画面覆盖率达到阈值，且目标仍完整入区。

B. Visual Completion and Milestone Review

固定前视相机作为日常任务的主要评价视角，侧视用于辅助人工审核遮挡或边界不清的情形。完整目标区域由独立参考轮廓定义，不使用被目标遮挡后的可见区域分割代替。目标暂时不可见时，不以可见掩码为空推断完整入区；无法可靠判断的时段不计为已确认成功。完成时目标须仍位于区域内，曾经入区后又被拨出不计为成功。表 X定义过程评价里程碑，不要求策略具有显式阶段输出。

若 N_j 为按任务顺序完成前 j 个里程碑的试验数、N 为全部试验数，阶段累计到达率与条件完成率分别为 N_j/N 与 N_j/N_{j-1}，其中 $N_0 = N$。分母为零时记为不适用，反复进入同一阶段不重复计数。各方法计时起点一致，完成时间取首次满足完整成功条件的时刻；若采用持续确认，则以确认结束时刻计时，并记录首次满足时刻。

（填写独立参考轮廓的获取方法、入区边界容差、可见性阈值、邻接距离及持续确认时间。如允许边界容差，在正文同步说明；不要以较宽松的训练阶段伪标签阈值替代最终成功判据。补充审核人数、分歧复核和仍无法判定试验的报告方式。）

C. Trial Allocation and Accounting

示范初始位置为 $(0,0)$、$(0,3)$、$(1,1)$、$(1,2)$、$(2,0)$ 和 $(2,3)$，基础测试覆盖全部 12 个网格位置与三类已见物体的组合，扩展测试使用未见颜色的布料。各方法使用相同的物体–位置–颜色组合和初始布置规则，逐试验记录物体、位置、布料颜色、checkpoint、训练种子、执行重复、完成时间和主要失败原因。总体结果与分组结果使用同一组试验，逐位置结果保留分母。整体成功率为完整测试池中的成功次数除以试验总数，不对相互重叠的分组列求和。位置与颜色分组比较匹配其余条件，区分未见位置、未见颜色及两者同时变化的组合。测试前冻结策略与分割模型，并核对未见位置和颜色是否出现在分割开发数据中。

（填写未见颜色、扩展测试组合及各组分母；若分割开发已使用这些条件，明确其仅对策略示范未见。更换颜色时尽量保持布料材质、尺寸及厚度一致。）

关键比较使用多个独立训练种子，区间估计保留训练种子与条件区组结构；同一 checkpoint 的执行重复不代替训练重复。每次试验记录一个主要失败类型，过程异常可另加标签，但不重复计入主要失败分布。

（核对示范的物体–位置清单与各条件样本数；补充实际重复次数、训练种子、区间估计方法、网格间距及布料 / 物体朝向重置规则。基础测试数为 36 乘以每个条件重复次数，扩展测试按预先固定的组合清单另计；消融沿用基础测试时与包含扩展条件的整体结果分开报告。预先规定超时、安全停止、技术故障的统计与重试方式，保留全部尝试记录。失败分类优先采用可审核的未完成里程碑或现象，不将推测的模型原因当作评价标签。）

D. Natural Error and Recovery Review

从各组全部执行视频与日志中识别自然错误，不仅抽取成功恢复的片段。预先固定下列可观察条件，按任务阶段解释运动与进展，不以模型自身的预测判定恢复：

- 布料缠阻：布料卷入或持续阻碍工具 / 目标运动；解除阻碍并恢复当前操作进展时记为恢复。
- 无效接触：工具完成面向目标的推动动作而未建立有效接触或产生目标进展；随后有效接触并推动目标时记为恢复。
- 输送偏差：目标被推离通往区域的操作路径或停在区域边缘未完整入区；重新建立朝向区域的有效输送时记为恢复，完整入区另记为里程碑。
- 执行停滞：尚未完成任务时，在预定持续时间内无有效进展；重新产生当前阶段的有效进展时记为恢复。

错误事件从首次满足判据时开始，到恢复、任务终止或预定恢复窗口结束时结束。窗口须有完整随访时间且恢复须持续达到确认时长；窗口内任务终止且未恢复的事件记为未恢复。同一连续事件不逐帧重复计数，只有完成恢复后再次出错才记录新事件。重叠现象采用一个主要事件类别，其他现象保留为辅助标签；持续输送偏差继而停滞等连续过程不拆成多个独立恢复机会。发生至少一次错误的试验计入错误发生率；恢复耗时仅在已恢复事件上计算，同时报告恢复数 / 事件数及出错试验的最终成功数 / 试验数。无事件时恢复率不适用；不可审核事件单列数量，不静默剔除。

（确定各阶段的有效进展判据、位移 / 接触容差、停滞时长、恢复窗口、确认时长及事件分类优先级。按物体、位置与错误类型报告样本量，必要时按严重程度分组；区间估计以试验为聚类单位。无法进行完整随访且未终止的事件单独报告。选取典型恢复、未恢复与有害修正片段，避免只展示成功案例。）

E. Representation Diagnostics

专家动作重构探针比较三种输入：上下文、审核描述量加上下文、预测描述量加上下文。使用容量匹配的小型读出模型和相同动作目标，训练与评价按完整轨迹分开；上下文的内容保持一致，不包含待预测专家动作或其未来观测。评价集同时记录描述量误差、标签有效比例和动作重构误差。该探针检验任务信息与动作选择的联系，不能替代部署侧专家标注，也不直接估计覆盖常数或条件误差差异。

（填写上下文、动作尺度、读出模型和独立评价轨迹。若历史模型使用速度或上一指令，列明其数据来源与时间范围，确保比较不引入未来信息。）

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